MILITARY SPECIFICATION

FUZE, POINT DETONATING, (PD), M759

This specification is approved for use by the US Army Armament, Munitions and Chemical Center and is available for use for all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the requirements for manufacture and acceptance of parts, assembly and packing for the M759, Point Detonating (PD) Fuze.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitations, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-Q-9858	-	Quality Program Requirements
MIL-R-398	-	Explosive, RDX
MIL-L-46225	-	Explosive, Lead Azide
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-E-81111	-	Explosive, PBXN-5

STANDARDS

MILITARY

MIL-STD-1167	-	Ammunition Data Card
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests for

FSC: 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

11834312	-	Fuze, PD, M759
11834336	-	Fuze Outer Packing
11834337	-	Fuze Inner Packing

PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Officer, Washington, DC 20402. Orders should specify, "49 CFR100-199 (latest revision)")

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications (see 4.5.1).

3.1.1 Fuze. The fuze assembly shall comply with all the requirements specified on drawing 11834312, all associated drawings, and with all requirements in applicable specifications and standards.

3.1.2 Government furnished property list. The Government furnished property list shall consist of the following:

a. Explosive, RDX, Type B per MIL-R-398
b. Explosive, Lead Azide per MIL-L-46225
c. Explosive, PBXN-5, Type I & II per MIL-E-81111

3.2 Components and assemblies.

3.2.1. Ballistic performance.

3.2.1.1 Premature burst. The fuze shall not function in bore within 15 meters of the gun muzzle during uninterrupted projectile flight.

3.2.1.2 Early burst. The fuze shall not function between 15 and 3000 meters of the gun muzzle during uninterrupted projectile flight.

3.2.1.3 Non-arming. The fuze shall not function upon impact with a 0.063 inch 2024-T3 aluminum plate at 0° angle of obliquity at 1.5 meters or less from the muzzle of the gun at a reliability rate of 95%. (See 4.6.1).

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3.2.1.4 Arming. The fuze shall function upon impact with a 0.063 inch 2024-T3 aluminum plate at 0° angle of obliquity at 150 ± 1 meters with a reliability of 90%. (See 4.6.2).

3.2.1.5 Target reliability. The fuze shall function upon impact with a 0.060 inch mild steel plate at 0° obliquity at a distance of 200 ± 2 meters with a reliability of 95%. (See 4.6.3).

3.2.1.6 Extended range sensitivity. The fuze shall function upon impact with supportive terrain from 3000 to 4000 meters at impact angle less than 80° obliquity with a reliability of 95%. (See 4.6.4).

3.2.1.7 Armor plate sensitivity. The fuze shall function upon impact with steel armor plate at 0° to 50° angle of obliquity at 500 to 2500 meters with fuze function reliability of 95%. (See 4.6.5).

3.2.2 Environmental and safety performance.

3.2.2.1 Jolt. The fuze shall withstand a 1/2 sine wave pulse of 230 ± 30 g's for a 2.0 ± 0.2 millisecond duration at a maximum pulse rate of 40 shocks/minute when tested in accordance with MIL--STD-331, Test 101.2. Fuzes subjected to this test are not required to be operable but must remain safe to store, transport and handle. (See 4.5.2.1.).

3.2.2.2 Jumble. The fuze shall withstand rough handling when tested in accordance with MIL-STD-331, Test 102.1. Fuzes subjected to this test are not required to be operable but must remain safe to store, transport and handle (See 4.5.2.2).

3.2.2.3 Salt fog. After exposure to a 5% salt solution for minimum of 48 hours in accordance with MIL-STD-331, Test 107.1, the fuze shall be safe to handle and meet the arming, non-arming and target reliability requirements. (See 4.5.2.3).

3.2.2.4 Transportation vibration/temperature. After undergoing a 5 to 500 hertz (HZ) continuous logarithmic sweep vibration with 0.50 inch peak to peak displacement at temperatures from -54° to 71°C in accordance with MIL-STD-331, Test 119, the fuze shall be safe to handle and meet the arming, non-arming and target reliability requirements. (See 4.5.2.4).

3.2.2.5 Waterproofness. The fuze shall withstand immersion for one hour when tested in accordance with MIL-STD-331, Test No. 108. Fuzes subjected to this test shall be safe to handle and meet the arming, non-arming and target reliability requirements. There shall be no evidence that any water has entered the fuze. (See 4.5.2.5).

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3.2.2.6 Temperature. The fuze shall operate properly within the temperature range of -54°C to 71°C.

3.3 Manufacturing process. All parts and assemblies shall be manufactured by the identical process used to produce the first article sample and no deviations from the process shall be made without prior Contracting Officer approval.

3.4 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. All parts and assemblies shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part or assembly, nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing material shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of-this detail specification.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items in sample quantities as indicated.

PART DESCRIPTION	DRAWING	QUANTITY
Fuze, 30mm, PD M759	11834312	1006

4.3.2 Inspection to be performed. See MIL-A-48078 and Table 1 included herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-70435(AR)

PARAGRAPH	TITLE		SHEET 1 OF 5		DRAWING NUMBER 11834312
	Fuze, 30mm, PD M759				NEXT HIGHER ASSEMBLY 11825925/11825948
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Fuze Assembly				
	Examination for defects	32	100%	3.1.1	4.4.2.10 (Critical 1, Major 101-110)
	Jolt	9		3.2.2.1	4.5.2.1
	Jumble	8		3.2.2.2	4..2.2
	Salt fog	36		3.2.2.3	4.5.2.3
	Transportation vibration temperature (TV-T)	921*		3.2.2.4	4.5.2.4 (Procedure I)
	Waterproofness	40		3.2.2.5	4.5.2.5
	Non-arming -54° ± 3°C	94	Note2,3	3.2.1.3	4.6.1
	21° ± 3°C	94			
	71° ± 3°C	94			
	Arming -54° ± 3°C	39	Note2,4	3.2.1.4	4.6.2
	21° ± 3°C	39			
	71° ± 3°C	39			
	Target reliability	378	Note2,5	3.2.1.5	4.6.3
	Extended range sensitivity	52	Note2,6	3.2.1.6	
	Armor plate sensivitity	52	Note2,6	3.2.1.7	4.6.5

NOTES:

*Additional units will be conditioned to replace "no test" situations. TV-T sample is used in subsequent ballistic tests and waterproofness tests.

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TABLE I · FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-70435(AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
	Fuze, 30mm, PD M759		2 OF 5		11834312
					NEXT HIGHER ASSEMBLY
					11825925/11825948
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD

NOTES

1. Sample size for examination shall be in accordance with applicable paragraph.
2. Reject if a single premature or early burst occurs during testing.
3. Accept on 9 or less defects, reject on 10 or more defects (non-arming). The accept/reject quantity is cumulative over all three temperatures, however no more than 7 defects are allowed at any given temperature.
4. Accept on 7 or less defects, reject on 8 or more defects (arming). The accept/reject quantity is cumulative over all three temperatures, however no more than 6 defects are allowed at any given temperature.

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

MIL-F-70435(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 5		DRAWING NUMBER
	Fuze, 30mm, PD M759				11834312
					NEXT HIGHER ASSEMBLY
					11825925/11825948
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
5.	Accept on 13 or less defects, reject on 14 or more defects (target reliability)				
6.	Accept on 3 or less defects, reject on 4 or more defects (extended range and armor sensitivity).				

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-70435(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Fuze, 30mm, PD M759	4 OF 5	11834312
			NEXT HIGHER ASSEMBLY 11825925/11825948

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Base (Dwg 11834310) Examination for defects	32		3.3.1	4.2.2.1
	Probe (Dwg 11834329) Examination for defects	32		3.3.1	4.2.2.2
	Pin, firing (less lubrication) (Dwg 11820627) Examination for defects	32		3.3.1	4.2.2.3
	Ogive (Dwg 11834331) Examination for defects	32		3.3.1	4.2.2.4
	Spring, piston (Dwg 11834326) Examination for defects	32		3.3.1	4.2.2.5

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

MIL-F-70435(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Fuze, 30mm, PD M759	5 OF 5	11834312
			NEXT HIGHER ASSEMBLY: 11825925/11825948

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Piston assembly (Dwg 11834313) Examination for defects	32		3.3.1	4.2.2.6
	Rotor assembly (Dwg 11834327) Examination for defects	32		3.3.1	4.2.2.7
	Body assembly (Dwg 11834328) Examination for defects	32		3.3.1	4.2.2.8
	Holder and body assembly (Dwg 11834315) Examination for defects	32		3.3.1	4.2.2.9

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of detonators shall contain:

a. parts bearing the same lot interfix number from one manufacturer.

b. Lead azide of the same type from not more than two lots produced by the same manufacturer, in accordance with the same specification and same specification revision which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

c. RDX from not more than one lot.

d. Primer mixture consisting of batches blended by the same process. Each batch shall be composed of the same ingredients, each of which is to be from one interfix lot from one manufacturer.

4.4.2 Examination. (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-70435(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Base	1 OF 1	11834310
			NEXT HIGHER ASSEMBLY 11834311

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Inside diameter of crimp flange (max)		100%	3.1.1	Gage/Auto gage
102.	Outside diameter of crimp flange (min)		100%	3.3.1	Gage/Auto gage
103.	Height of crimp flange (min)		100%	3.3.1	Gage/Auto gage
104.	Major diameter of threads	4.4.2	.40%	3.3.1	Gage
MINOR:					
201.	O.D. Fully threaded	4.4.2	.65%	3.3.1	Visual/Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

MIL-F-70435(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.	Probe	1 OF 1	11834329
			NEXT HIGHER ASSEMBLY 11834312

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR:					
101.	Diameter of datum A	4.4.2	.40%	3.3.1	Gage
102.	Length from shoulder to rear face		.40%	3.3.1	Gage
MINOR:	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Pin, Firing (Less Lubrication)	1 OF 1	11820627
			NEXT HIGHER ASSEMBLY 11834332

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Profile of point	4.4.2	.40%	3.3.1	Gage
102.	Surface finish of shaft		.40%	3.3.1	Visual
MINOR:					
201.	Overall length	4.4.2	.65%	3.3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.1	Ogive		11834331
			NEXT HIGHER ASSEMBLY
			11834312

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Diameter of holder and body assembly cavity	4.4.2	.40%	3.3.1	Gage
102.	True position of hold and body assembly cavity		Note 1	3.3.1	Gage
103.	Length from base stop to holder assembly stop		.40%	3.3.1	Gage
104.	Diameter of piston assembly cavity		.40%	3.3.1	Gage
105.	Length from base stop to piston assembly stop		.40%	3.3.1	Gage
106.	Diameter of probe cavity		.40%	3.3.1	Gage
107.	Length from base stop to front face		.40%	3.3.1	Gage
108.	Pitch diameter of external threads		.40%	3.3.1	Gage
109.	Location of ball groove ramp		Note 1	3.3.1	Gage
110.	Angle of ball groove ramp		Note 1	3.3.1	Gage
111.	Presence of lubricant on piston assembly cavity		.40%	3.3.1	Visual
112.	Minor diameter of internal threads		.40%	3.3.1	Gage

NOTES:

1. n = 10, c = 0 per machine when this is a tool controlled feature (5 at start and 5 at finish of each shift). If feature is not tool controlled, the 0.40% AQL applies.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.4	Ogive		11834331
			NEXT HIGHER ASSEMBLY 11834312

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR:					
201.	Surface finish (internal functional surfaces)	4.4.2	.65%	3.3.1	Visual/Gage
202.	Burrs (at transitions of probe, piston assembly or holder and body assembly cavities)		.65%	3.3.3	Visual
203.	Foreign matter (excluding lubrication)		.65%	3.3.3	Visual
204.	Internal chamfers or radii improper, missing or incorrect		.65%	3.3.3	Visual
205.	I.D. Surface fully threaded		.65%	3.3.3	Visual/Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

MIL-F-70435(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11834326
4.4.2.5	Spring, Piston				NEXT HIGHER ASSEMBLY 11834309
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Spring force (destructive load test)	50/lot	Acc 1 Rej 2	3.3.1	Gage
MINOR:					
201.	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Piston Assembly	1 OF 1	11834313
			NEXT HIGHER ASSEMBLY 11834312

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Length of firing pin in assembly	4.4.2	.40%	3.3.1	Gage
102.	True position of firing pin greater than 50% or tolerance		.40%	3.3.1	Gage
103.	Tears, nicks, cuts or voids in piston seal		.40%	3.3.1	Visual
104.	Lubrication on firing pin, piston base, or piston seal		.40%	3.3.1	Visual/Manual
105.	Flow rate check	Note 1	100%	3.3.1	Automatic Gage
106	Firing pin pushout test (destructive test)	40	Note 2	3.3.1	Gage
MINOR:					
201.	Foreign material except lubricant	4.4.2	.65%	3.3.3	Visual
202.	True position of firing pin less than or equal to 50% over tolorance.		.65%	3.3.1	Gage

NOTES:

1. This check is performed 100% on a qualified and verified assembly machine, with defects being automatically removed. If the machine fails verification or if the assemblies are hand assemblies are 100% checked (bench or machine) with defects removed and then submitted for inspection verification of check.
2. Reject on one or more defects

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Rotor Assembly	1 OF 1	11834327
			NEXT HIGHER ASSEMBLY 11834328

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Presence of detonator assembly	4.4.2	.40%	3.3.1	Visual
102.	Detonator assembly inverted		.40%	3.3.1	Visual
103.	Presence of rotor sear		.40%	3.3.1	Visual
104.	Width of firing pin slot (min)		.40%	3.3.1	Gage
105.	Detonator pushout test (destructive test)	40/lot	Acc 1	3.3.1	Gage
106.	Detonator assembly damaged (Note 1)		.40%	3.3.1	Visual
107.	Rotor sear damaged or incomplete (Note 1)		.40%	3.3.1	Visual
MINOR:					
201.	Foreign matter	4.4.2	.65%	3.3.3	Visual
202.	Metal defects, cracks, or splits		.65%	3.3.3	Visual
203.	Presence of lubrication		.65%	3.3.1	Visual

NOTES:

1. Visual defects may be used for determining acceptance of this characteristic if approved by the procuring agency.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Body Assembly		11834328
			NEXT HIGHER ASSEMBLY 11834309

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Arm/non-arm spin check	Note 1	100%	3.3.1	Gage
MINOR:					
201.	Presence of roller weight and lockweight assembly	4.4.2	.65%	3.3.1	Visual/X-ray

NOTES:
This check is performed 100% on a qualified and verified assembly machine, with the defects being automatically removed. If the machine fails verification or if the assemblies are hand assembled, the assemblies are 100% checked (machine or bench) with defects removed and then submitted for inspection verification of check.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Holder and Body Assembly	1 OF 1	11834318
			NEXT HIGHER ASSEMBLY 11834312

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Outside diameter (max)	4.4.2	.40%	3.3.1	Gage
102.	Length from rear face of body holder to piston assembly bearing surface of spring retainer		.40%	3.3.1	Gage
103.	Presence of piston spring, support washer, spring retainer, or less than three balls.		.40%	3.3.1	Visual
MINOR:					
201.	Foreign matter, excluding lubrication	4.4.2	.65%	3.3.3	Visual

NOTES:

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PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.4.2.10	Fuze, 30MM, PD M759		11834312
			NEXT HIGHER ASSEMBLY
			11835925/118325948

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:					
1.	Rotor assembly in the unsafe position	Note 1	100%	3.3.1	Gage/X-ray
MAJOR:					
101.	Height of spitback assembly retaining crimp (max)	4.4.2	100%	3.3.1	Auto/Gage
102.	Inside diameter of spitback assembly retaining crimp (max)		100%	3.3.1	Auto/Gage
103.	Pitch diameter of external thread		.40%	3.3.1	Gage
104.	External threads damaged		.40%	3.3.1	Visual
105.	Length from rear face of ogive to rear face of base (max)		.40%	3.3.1	Gage
106.	Length from rear face of ogive to top of probe cap (max)		.40%	3.3.1	Gage
107.	Reverse torque (min)	10	Note 2	3.3.1	Gage
108.	Presence of or improperly assembled probe, confinement cup or setback cup.		.40%	3.3.1	Visual/X-ray
109.	Presence sealing compound form base/ ogive joint		.40%	3.3.1	Visual

NOTES:

See sheet 3.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Fuze, 30MM, PD M759	2 OF 3	11834312
			NEXT HIGHER ASSEMBLY 11835925/118325948

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
110.	Probe cap missing or improperly crimped		.40%	3.3.1	Visual
111.	Jolt	9		3.2.2.1	4.5.2.1
112.	Jumble	8		3.2.2.2	4.5.2.2
113.	Transportation (TV-T) Note 1	807		3.2.2.4	4.5.2.4 (Procedure II)
114.	Waterproofness	40		3.2.2.5	4.5.2.5
115.	Non-arming -54° ± 3°C	90	Table 2	3.2.1.3	4.6.1
	21° ± 3°C	90			
	71° ± 3°C	90			
116.	Arming -54° ± 3°C	41	Table 2	3.2.1.4	4.6.2
	21° ± 3°C	41			
	71° ± 3°C	41			
117.	Target reliability	270	Table 2	3.2.1.5	4.6.3
118.	Extended range sensitivity	52	Table 2	3.2.1.6	
119.	Armor plate sensivitity	52	Table 2	3.2.1.7	4.6.5
MINOR:					
201.	Marking missing, incorrect, or illegible	4.4.2	.65%	3.3.3	Visual

NOTES:
1. TV-T sample is used in subsequent ballistic tests and waterproofness test.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 11834312
4.4.2.10	Fuze, 30MM, PD M759				NEXT HIGHER ASSEMBLY 11835925/118325948
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
NOTES 1.	This check is performed 100% on qualified asembly machine with the defects automatically removed. If machine fails verification, or if the assemblies are hand built, the assemblies are 100% x-ray checked with the critical defects manually removed and the lot then submitted for inspection verification of the x-ray sort.				
2.	Reject on one or more defects.				

NOTES:

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Table 2 Lot Acceptance Testing
Functional Acceptance Criteria

TEST PARAMETER	N_i initial test sample	C_i initial accept number	V_i initial reject number	N_2 second test sample	N_t total test sample	C_t total accept number	V_t total reject number	Notes
Non-arming	90	2	9	180	270	8	9	1,2,4,5
Arming	39	2	9	84	123	8	9	1,2,4,5
Target Reliability	90	2	11	180	270	10	11	1,3,4,5
Extended range	30	1	4	22	52	3	4	1,3,4,5
Armor plate	30	1	4	22	52	3	4	1,3,4,5

Note 1: For a given test parameter, select and test the initial test sample (N_i). If the number of defects is equal to, or less than, the initial acceptance number (C_i) accept the test. If the number of defects equals or exceeds the initial reject number (V_i) reject the test. If the number of defects is greater than C_i but less than V_i select a second test sample (N_2) and subject them to the test conditions. Accept the test if the total number of defects (cumulative for N_i & N_2) is equal to, or less than the total acceptance number (C_t). Reject the test if the total number of defects equals or exceeds the total reject number (V_t).

Note 2: Test samples (N_iN_i) will be devided into three test sub groups. One subgroup willbe conditioned and tested A -54°C, the second subgroup at +21°C, and the third at +71°C. Failures will be accumulated across the temperature subgroups and evaluated accordingly.

Note 3: Test sample will be conditioned & tested at ambient (+21°C).

Note 4: The lot shall be rejected if a signel premature or early burst occurs during ballistic test.

Note 5: All fuzes undergoing ballistic testing shall be pre-conditioned in accordance with 4.5.2.6 and 4.6. In order to assure the required sample is available for testing, units may be pre-conditioned without being ballistically tested. These unused pre-conditioned LAT units may be returned to the lot for delivery.

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Note 6: The target reliability test is not required if the arming test $C_i=0$. If $C_i=1$, an additional sample of 21 fuzes may be tested for arming, 7 at each temperature. If no additional failures occur ($C_t=1$ for $N_t=60$) the target reliability test is not required. For arming C_t of 2 or more, the target reliability test is required.

Note 7: Test parameter is required only when specified in the contract or purchase order.

4.4.3 Testing. Testing is described in the first article table and quality conformance inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the "Paragraph Reference/Inspection Method" column starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6.2 of MIL-A-48078 and 6.4 herein.

4.4.5 Deterioration of explosive components.

4.4.5.1 Check test for deterioration of detonators. If the total time between original acceptance of any lot and the assembly of that lot into the fuze exceeds two years, or if the detonators have been subjected to adverse conditions at any time since previous tests, the detonator lot shall be subjected to and must satisfactorily pass the functional tests specified on the drawing prior to assembly into the fuze. The sampling plan for these tests shall be in accordance with MIL-STD-105D, Normal Inspection, to an AQL of 0.40 percent. This test shall be performed by the contractor on detonators selected by inspection at the facility assembling the detonator into the fuze.

4.4.5.2 Check test for deterioration of spitback lead assemblies. If the total time between original acceptance of any lot and the assembly of that lot into the fuze exceeds two years, or if the lead cup assemblies have been subjected to adverse conditions at any time since previous tests, the lead cup lot shall be subjected to and must satisfactorily pass the function test specified on the applicable drawing prior to assembly into the fuze. The sampling plan for these tests shall be in accordance with MIL-STD-105D, Normal Inspection, to an AQL of 0.40 percent. This test shall be performed by the contractor on lead cup selected for inspection at the facility assembling the lead cup into the fuze.

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4.5 Test methods and procedures.

PRECAUTION - This specification covers sampling and testing of explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Section 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Environmental and safety test procedure. The component parts of the fuze, which have been disassembled and inspected following the test, shall be scrapped.

4.5.2.1 Jolt. The test and inspection shall be performed in accordance with the procedures specified in MIL-STD-331, Test No. 101.2 to demonstrated compliance with the requirement of 3,2.1.1.

4.5.2.2 Jumble. The test and inspection shall be performed in accordance with the procedures specified in MIL-STD-331, Test No. 101.2 to demonstrate compliance with the requirement of 3.2.1.2.

4.5.2.3 Salt fog. The test and inspection shall be performed in accordance with the procedures specified in MIL-STD-331, Test No. 107.1. The fuzes shall be subjected to the 48 hour test only. All test items shall be disassembled and inspected as specified in Test No. 107.1 to demonstrate compliance with the requirement of 3.2.2.3.

4.5.2.4 Transportation vibration temperature The test and inspection shall be performed in accordance with the procedures specified in MIL-STD-331, Test No. 119. Fuzes subjected to the waterproof test of 4.5.2.5 shall be preconditioned by this test and fuzes subjected to ballistic testing shall be preconditioned by this test prior to being tested in accordance with 4.6.1 through 4.6.5.

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4.5.2.5 Waterproofness. The test and inspection shall be performed in accordance with the procedures specified in MIL-STD-331, Test No. 108. The water should contain a dye penetrant so leakage will be easily detected when fuzes are disassembled. All fuzes will be disassembled to determine compliance with 3.2.2.5.

4.5.2.6 Temperature. For ballistic test requiring temperature extreme conditioning, the fuzes assembled to cartridge shall be conditioned for six hours minimum at the required temperature.

4.6 Ballistic test procedure. As specified in the contract, the fuze shall be assembled to either an M789 HEDP Cartridge or equivalent, and fired in the M230E1 auto gun or in a 6 1/2 degree constant twist Mann barrel conforming to drawing 28200380. Unless otherwise specified, gun and ammunition are to be fired at existing atmosphere conditions.

4.6.1 Non-arming. The fuzes assembled to cartridges as specified in the contract shall within two (2.0) minutes after removal from the temperature conditioning chamber, to be tested at a range of 15.0 ± 0.5 meters with an angle of impact of 0° ± 10° on 0.063 inch 2024-T3 aluminum.

4.6.2 Arming. The fuzes assembled to cartridges as specified in the contract, shall within two (2) minutes after removal from the temperature conditioning chamber be tested at a range of 15.0 ± 0.5 meters with an angle of impact of 0° ± 10° on 0.063 inch 2024-T3 aluminum.

4.6.3 Target reliability. The fuzes assembled to cartridges as specified in the contract shall within two (2) minutes after removal from the temperature conditioning chamber be tested at a range of 200 ± 2 meters with an angle of impact of 0° ± 10° on 0.60 inch (16 gage) mild steel plate.

4.6.4 Extended range sensitivity. The fuzes assembled to cartridges as specified in the contract shall within two (2) minutes after removal from the temperature conditioning chamber be tested at a range of 3000 to 4000 meters impact on available supportive terrain and a impact angle less than 800 obliquity. See paragraph 6.3.6 for definition of acceptable terrain.

4.6.5 Armor plate sensitivity. The fuzes assembled to HEPD cartridges shall be tested at a range of 500 ± 25 meters with an angle of impact of 50° - 5° on 0.50 inch minimum steel.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and requirements. Packaging shall be as specified on Drawing 11834337and Department of Transportation Regulation 49 CFR 100-199.

5.2 Packing and marking. Packing and marking shall be as specified on Drawing 11834336

6. DEFINITIONS AND NOTES

6.1 Intended use. The fuzes are intented for use in the M230E1, DEFA 553 and ADEN MKIV 30MM automatic weapon systems when assembled to the 30MM High Explosive Dual Purpose Cartridge.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Title, number and date of this specification.

6.2.2 Place of inspection if not at the place of manufacture.

6.2.3 Provision for the supply, maintenance, and disposition of mandatory test equipment for acceptance inspection process.

6.2.4 Instructions for shipment of production lot samples for functional tests.

6.2.5 Provision for preparation for delivery if required for indefinite storage interplant shipment of the lead assembles.

6.2.6 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-Q-9858, Quality Program Requirements.

6.2.7 provisions for MIL-STD-1167, Ammunition Data Cards, on DD form 1423, Contract Data Requirements List.

6.2.8 Provisions for the inclusion of Test Major 118 and 119 of paragraph 4.2.2

6.3 Definitions.

6.3.1 Safe fuze. The criteria by which fuzes are judged to be safe are that:

a. no element shall explosively react

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b. no parts shall be broken, deformed, displaced, come apart, or become positioned in such a manner as to make the assembly unsafe to handle or dangerous.

c. if damaged or irregularity does not prevent assembly of the fuze to the cartridge, the fuze must be safe if used without knowledge of the damage.

6.3.2 Range. The distance between the muzzle of the gun and the target plate.

6.3.3 Non-armed. The rotor is locked in the position in which the rotor slot, normally occupied by the firing pin, is aligned with the upper body firing pin hole.

6.3.4 Armed fuze. Any fuze that is other than non-armed is considered armed.

6.3.5 No-test. A ballistic test round which failed to function because it missed the target or struck the target plate so close to the outside edge or the edge of a previous hole such that the full diameter of the fuze did not contact the plate material.

6.3.6 Acceptable terrain. Any supportive soil or surface such as clay, sand, sod, small rock, concrete, macadam, etc. is an acceptable impact media. Non-supportive soil or surface such as marsh, water, or snow is considered an unacceptable impact media. any fuze firing defect found to have been caused by an unacceptable impact media shall be considered a no-test.

6.3.7 Defective weapon. IF any fuze firing defect is considered to have been caused by a defective test weapon, and verified by the government, the firing defect shall be considered a no-test. In such cases, the defective test weapon shall be corrected or replaced and the test repeated in a whole or in part, as indicated.

6.4 Submission of inspection equiment for design approvals. See MIL-A-48078. Submit 2 copies of designs as required to amccom, attn; DRSMC-QAF-S (D), Dover, NJ 07801

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6.5 Drawings. Drawings listed in Section 2 of this specification under the heading of US Armament, Munitions and Chemical Command, (AMCCOM), may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of AMCCOM.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-A929)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-70435	FUZE, POINT DETONATING, (PD), M759

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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